THE CURIOSITY CUP 2025
A Global SAS® Student Competition

Superstore Marketing Campaign: Predicting Gold Membership Acceptance of Customers

IShowSAS

Abstract

This report presents an analysis of a superstore's marketing strategy, focusing on the effectiveness of a campaign offering gold memberships, which provide a 20% discount on all purchases. Using predictive modelling techniques, customer behaviours and preferences were analysed to determine the key factors influencing offer acceptance. The study leverages data from past campaigns and applies machine learning techniques such as logistic regression, decision trees, and neural networks using SAS Enterprise Miner. Neural Network is the most accurate predictive model and can be integrated into the superstore's marketing platform to identify customers likely to accept offers to increase the chances of successfully selling gold memberships.

Introduction

A superstore is planning for the year-end sale and is considering a promotional campaign featuring a gold membership, entitling customers to a 20% discount on all purchases, for only $499 which is $999 on other days. This exclusive deal is targeted solely at existing customers, with plans to contact them via phone. To maintain its competitiveness, the superstore has to look for ways to improve its marketing strategies and customer engagement by making use of the dataset of the customers' responses to accepting the membership offers with data, such as the financial metrics and the activities of customers on last year's campaign and building predictive models. The management is exploring the idea of creating a predictive model to identify potential customers likely to accept the offer. The primary goal is to identify the customer segments most likely to accept the offer, thereby increasing the chances of successfully promoting gold memberships and improving the efficiency of marketing resource allocation.

Predictive modelling is employed to analyse customer responses from a prior campaign, considering financial metrics, past shopping behaviour, and demographic attributes. Advanced analytics and predictive models, including decision trees, logistic regression, and neural networks, are utilised to uncover key predictors of customer acceptance. Recent studies show that predictive modelling in marketing campaigns is a well-established method to enhance targeting and efficiency, allowing businesses to allocate resources more effectively and increase the return on investment (García-Laencina et al., 2015). By leveraging advanced analytics techniques, the superstore aims to enhance its marketing efforts, improve customer targeting, and ultimately increase the acceptance rate of the promotional offer, driving higher sales and customer loyalty (Wedel & Kannan, 2016).

Exploratory Data Analysis

Columns: ☐ Label ☐ Mining ☐ Basic ☐ Statistics

Name	Role	Level	Report	Order	Drop	Lower Limit	Upper Limit
Complain	Input	Binary	No		No	.	.
Dt_Customer	Time ID	Interval	No		No	.	.
Education	Input	Nominal	No		No	.	.
Id	ID	Nominal	No		No	.	.
Income	Input	Interval	No		No	.	.
Kidhome	Input	Interval	No		No	.	.
Marital_Status	Input	Nominal	No		No	.	.
MntFishProduc	Input	Interval	No		No	.	.
MntFruits	Input	Interval	No		No	.	.
MntGoldProds	Input	Interval	No		No	.	.
MntMeatProdu	Input	Interval	No		No	.	.
MntSweetProd	Input	Interval	No		No	.	.
MntWines	Input	Interval	No		No	.	.
NumCatalogPu	Input	Interval	No		No	.	.
NumDealsPurcl	Input	Interval	No		No	.	.
NumStorePurcl	Input	Interval	No		No	.	.
NumWebPurch	Input	Interval	No		No	.	.
NumWebVisits	Input	Interval	No		No	.	.
Recency	Input	Interval	No		No	.	.
Response	Target	Binary	No		No	.	.
Teenhome	Input	Interval	No		No	.	.
Year_Birth	Input	Interval	No		No	.	.

Figure 1. Measurement Level

Since the objective is to identify which factors influence the decision of customers to accept the membership offer, the role of the response variable is set as the target. The role of the ID variable is not an input for this analysis so it is set as ID. The response variable contains data of either 1, indicating that the customer accepted the offer, or 0, indicating otherwise in the last year's campaign. The complain variable also contains data of either 1, indicating that the customer made a complaint, or 0, indicating no complaint has been made in the last 2 years. Therefore, the level of response and complain are set as binary. Inputs with character data type should have a nominal measurement level, while inputs with a numeric data type should have an interval measurement level.

Data Cleaning and Processing

Name	Method	Number of Bins	Role	Level
Complain	Default	4	Input	Binary
Education	Default	4	Input	Nominal
Income	Default	4	Rejected	Interval
Kidhome	Default	4	Input	Interval
Marital_Status	Default	4	Input	Nominal
MntFishProduc	Log	4	Input	Interval
MntFruits	Log	4	Input	Interval
MntGoldProds	Log	4	Input	Interval
MntMeatProdu	Log	4	Input	Interval
MntSweetProd	Log	4	Input	Interval
MntWines	Log	4	Input	Interval
NumCatalogPu	Log	4	Input	Interval
NumDealsPurcl	Log	4	Input	Interval
NumStorePurcl	Log	4	Input	Interval
NumWebPurch	Log	4	Input	Interval
NumWebVisits	Log	4	Input	Interval
REP_Income	Log	4	Input	Interval
Recency	Default	4	Input	Interval
Response	Default	4	Target	Binary
Teenhome	Default	4	Input	Interval
Year_Birth	Default	4	Input	Interval

Figure 2. Log Transformation

Based on observation, the Mnt, Num, and Income variables have skewness in their distributions. Therefore, in the variable edit tab, the method of these variables is changed from default to log so that these variables can undergo log transformation.

Data Partitioning

The dataset was partitioned into an 80-20 split, where 80% of the data was allocated to the training set and 20% to the validation set. This partitioning ensured that the model's performance could be accurately assessed on new, unseen data while reducing overfitting risks. The Data Partition node in SAS Enterprise Miner was used to implement this split.

Model Building

Fit Statistics

Target=Response Target Label=' '

Fit Statistics	Statistics Label	Train	Validation
NOBS	Sum of Frequencies	1790.00	450.000
MISC	Misclassification Rate	0.13	0.147
MAX	Maximum Absolute Error	0.92	1.000
SSE	Sum of Squared Errors	385.79	106.311
ASE	Average Squared Error	0.11	0.118
RASE	Root Average Squared Error	0.33	0.344
DIV	Divisor for ASE	3580.00	900.000
DFT	Total Degrees of Freedom	1790.00	.

Figure 3. Fit Statistics of Decision Tree

A Decision Tree model was applied to classify customer acceptance of the promotional offer. The splitting rule was based on the highest log-worth values computed from the Pearson Chi-square p-values for categorical variables and the F-test for interval variables. The tree depth was limited to six levels to prevent overfitting. The misclassification rates for the training and validation sets were 13.30% and 14.67%, respectively. Key predictors included income, recency of last purchase, and number of web visits.

Fit Statistics	Statistics Label	Train	Validation
AIC	Akaike's Information Criterion	1205.969	.
ASE	Average Squared Error	0.100329	0.101699
AVERR	Average Error Function	0.330718	0.32491
DFE	Degrees of Freedom for Error	1779	.
DFM	Model Degrees of Freedom	11	.
DFT	Total Degrees of Freedom	1790	.
DIV	Divisor for ASE	3580	900
ERR	Error Function	1183.969	292.4193
FPE	Final Prediction Error	0.10157	.
MAX	Maximum Absolute Error	0.987714	0.955183
MSE	Mean Square Error	0.10095	0.101699
NOBS	Sum of Frequencies	1790	450
NW	Number of Estimate Weights	11	.
RASE	Root Average Sum of Squares	0.316748	0.318902
RFPE	Root Final Prediction Error	0.318701	.
RMSE	Root Mean Squared Error	0.317726	0.318902
SBC	Schwarz's Bayesian Criterion	1266.359	.
SSE	Sum of Squared Errors	359.1789	91.52886
SUMW	Sum of Case Weights Times Fr...	3580	900
MISC	Misclassification Rate	0.137989	0.14

Figure 4. Fit Statistics of Logistic Regression

A logistic regression model was built using stepwise selection, where variables were included based on statistical significance. The selected predictors were web visit frequency, catalog purchases, teen household members, marital status, and recency of last purchase. The logit link function was applied to map binary outcomes, and the validation misclassification rate was 14.00%.

Fit Statistics	Statistics Label	Train	Validation
DFT	Total Degrees of Freedom	1790	.
DFE	Degrees of Freedom for Error	1753	.
DFM	Model Degrees of Freedom	37	.
NW	Number of Estimated Weights	37	.
AIC	Akaike's Information Criterion	1241.46	.
SBC	Schwarz's Bayesian Criterion	1444.589	.
ASE	Average Squared Error	0.098893	0.102035
MAX	Maximum Absolute Error	0.988788	0.96078
DIV	Divisor for ASE	3580	900
NOBS	Sum of Frequencies	1790	450
RASE	Root Average Squared Error	0.314472	0.319429
SSE	Sum of Squared Errors	354.0352	91.83116
SUMW	Sum of Case Weights Times Fr...	3580	900
FPE	Final Prediction Error	0.103067	.
MSE	Mean Squared Error	0.10098	0.102035
RFPE	Root Final Prediction Error	0.321041	.
RMSE	Root Mean Squared Error	0.317773	0.319429
AVERR	Average Error Function	0.326106	0.32224
ERR	Error Function	1167.46	290.0163
MISC	Misclassification Rate	0.126816	0.137778
WRONG	Number of Wrong Classifications	227	62

Figure 5. Fit Statistics of Neural Network

A neural network model was trained with three hidden layers to capture non-linear relationships. Initially, a three-hidden-unit configuration was tested, but increasing the number of hidden units to four improved model performance. The optimal iteration number was determined to be seven, balancing flexibility and complexity. The final misclassification rate was 13.78% for validation, making it the best-performing model.

Model Comparison

The Decision Tree, Logistic Regression, and Neural Network models were compared using misclassification rates and ROC curves. The Neural Network model had the highest area under the curve (AUC) and the lowest validation misclassification rate of 13.78%, outperforming the other models.

ROC

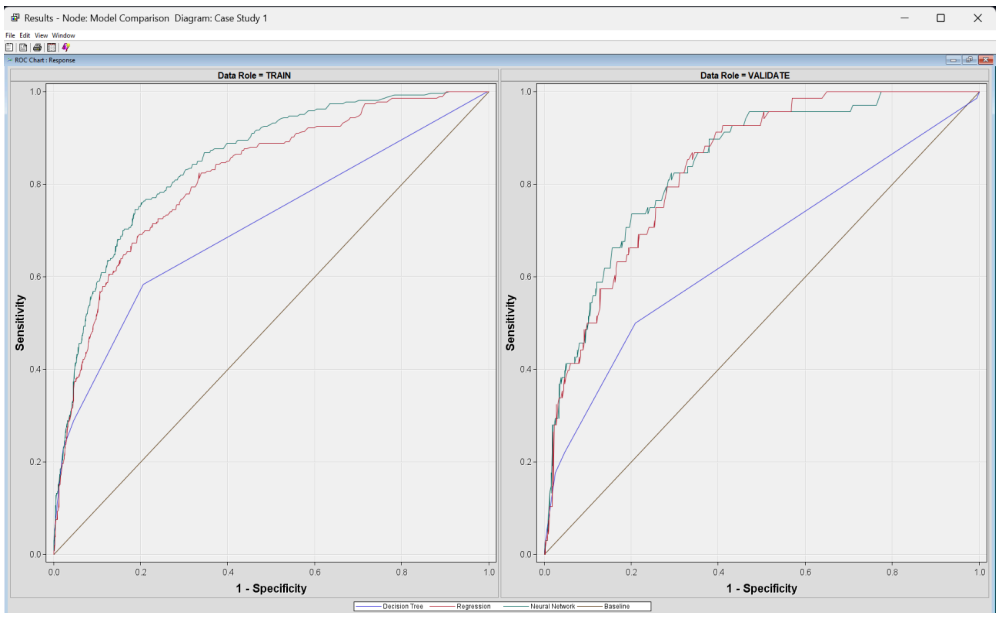

Figure 6. ROC Chart

Fit Statistic

Model	Misclassification Rate (Validation)	Accuracy
Decision Tree	14.67%	85.33%
Logistic Regression	14.00%	86.00%
Neural Network	13.78%	86.22%

Conclusion

The Neural Network model was selected as the final predictive model due to its lowest misclassification rate and highest predictive accuracy. Its ability to capture complex relationships between variables makes it the most suitable model for identifying potential customers likely to accept the promotional offer. Key variables such as web visit frequency, catalog purchases, and recency of purchase played a crucial role in the model's predictive power.

References

García-Laencina, P. J., Sancho-Gómez, J. L., Figueiras-Vidal, A. R., & Verleysen, M. (2015). Pattern classification with missing data: A review. Neural Computing and Applications, 19(2), 263-282.

Ahzan, R.(2023, January 2). *Superstore Marketing Campaign Dataset*. Kaggle. https://www.kaggle.com/datasets/ahsan81/superstore-marketing-campaign-dataset?resource=download

Wedel, M., & Kannan, P. K. (2016). Marketing analytics for data-rich environments. Journal of Marketing, 80(6), 97-121.

Appendix

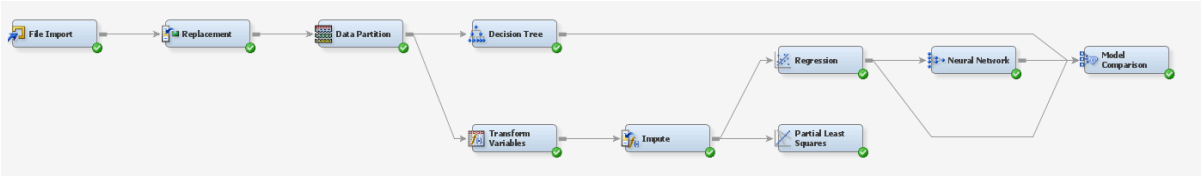

Figure 7: Full Data Mining and Predictive Modelling Diagram